

Boutique Winery Sales

Team 20 🚀



Business Question

What is the best way to target our customer base across priority segments, regions, and marketing channels to unlock the greatest customer value?



Data Set Overview

Data Overview

- **65,534 total sales** records analyzed, representing **22,869 unique customers**
- Data covers the period **2008–2010**

Sales Channels

- Email, Newsletter, Tasting Room, Winemaker Call

Customer Segments

- High Roller, Casual Visitor, Wine Enthusiast, Luxury Estate

Customer Location

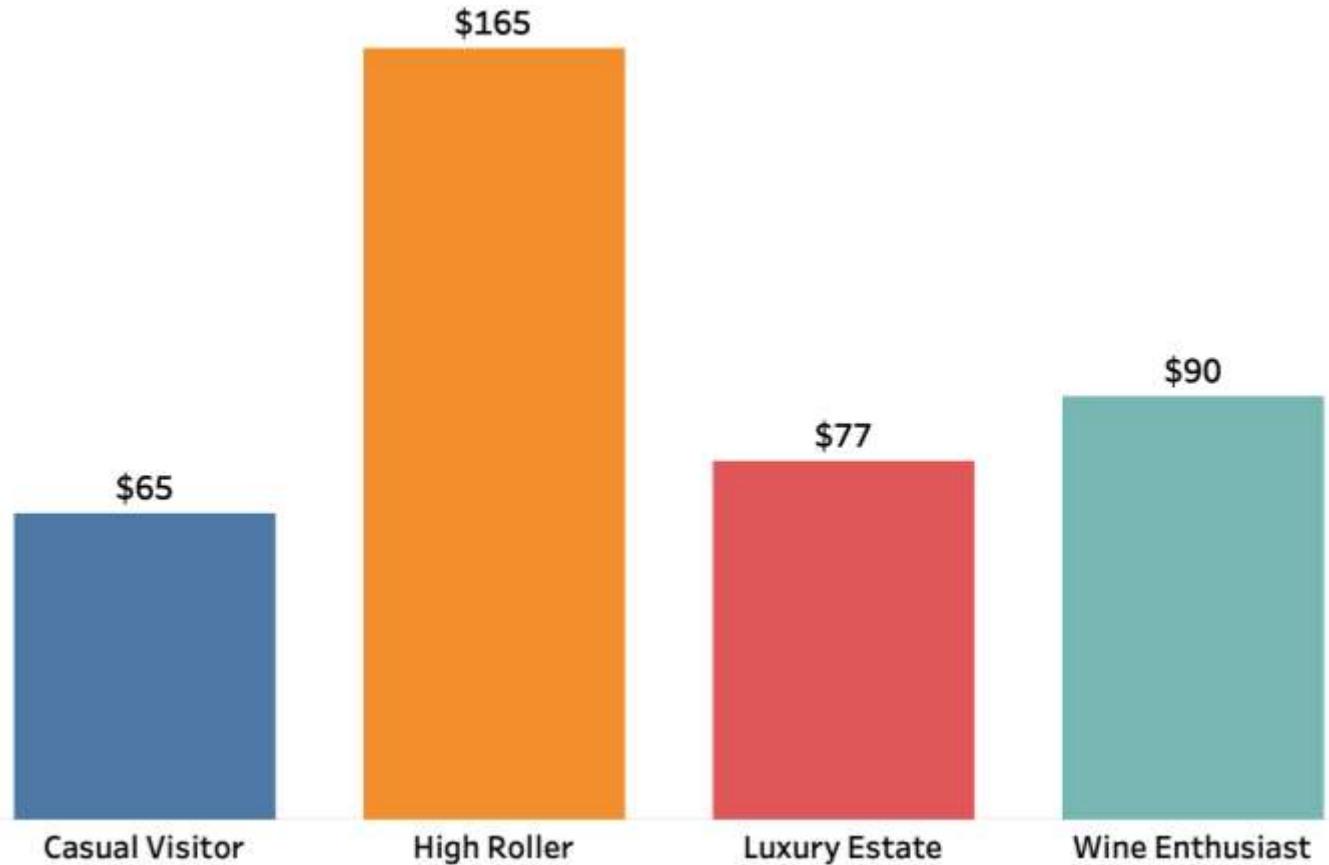
- ZIP Code, State

Tools Used

- Python – Pandas, NumPy, SciKit Learn
- Tableau



Average Sales by Customer Segment



Customers in the **High Roller** segment spend true to their category name.

Luxury Estates are still recovering from the '08 financial crisis.



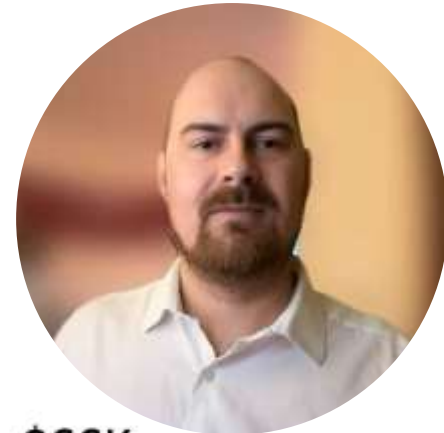
Sum of Sales by Customer Segment



The volume of **Casual Visitors** made them the highest spending customer segment with most of their purchases occurring at the Tasting Room.



Total Sales by Channel



Average Sales by Marketing Channels Per Subscriber Status

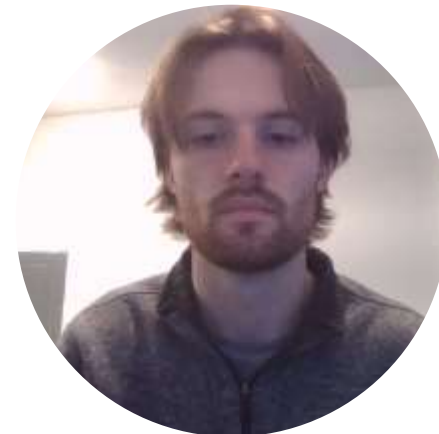
Channel	Subscribed	Not Subscribed
Email	\$104.88	\$78.24
Newsletter	\$89.48	\$86.65
Winemaker Call	\$96.84	\$84.14



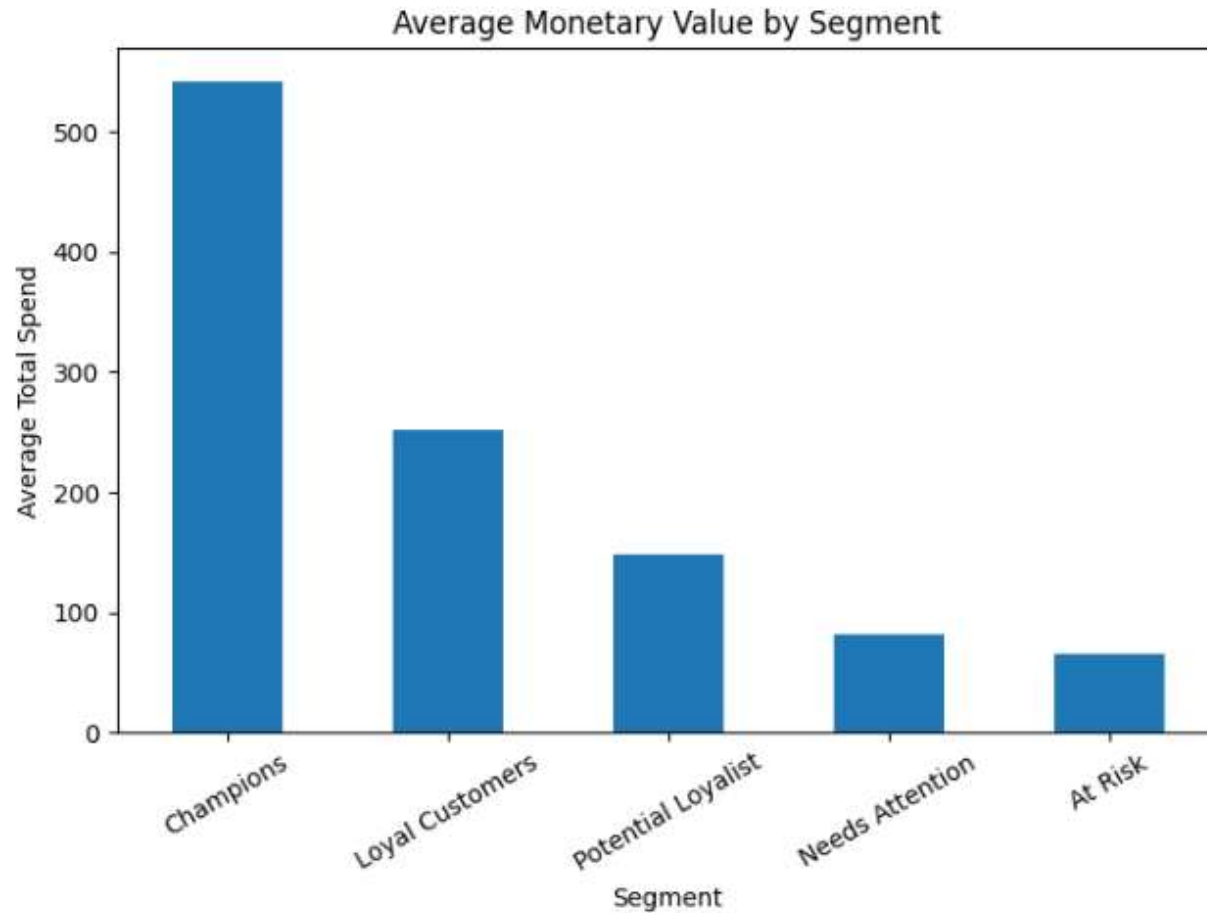
Recency, Frequency, Monetary Value (RFM) Analysis

- Calculated Recency, Frequency, and Monetary values for each customer by grouping data by Customer ID.
- Scored each metric from 1 to 4, using quartiles for Recency and a custom method for Frequency and Monetary to handle skewed data.
- Combined the three scores into a total RFM score.
- Used the total score to segment customers into five groups: Champions, Loyal Customers, Potential Loyalist, Needs Attention, and At Risk.

RFM Segments	Count	Average Sales
Champions	5775	\$114.61
Loyal Customers	4927	\$89.31
Potential Loyalist	5830	\$73.10
Needs Attention	6261	\$41.40
At Risk	76	\$62.09



Total Spend by RFM Segment

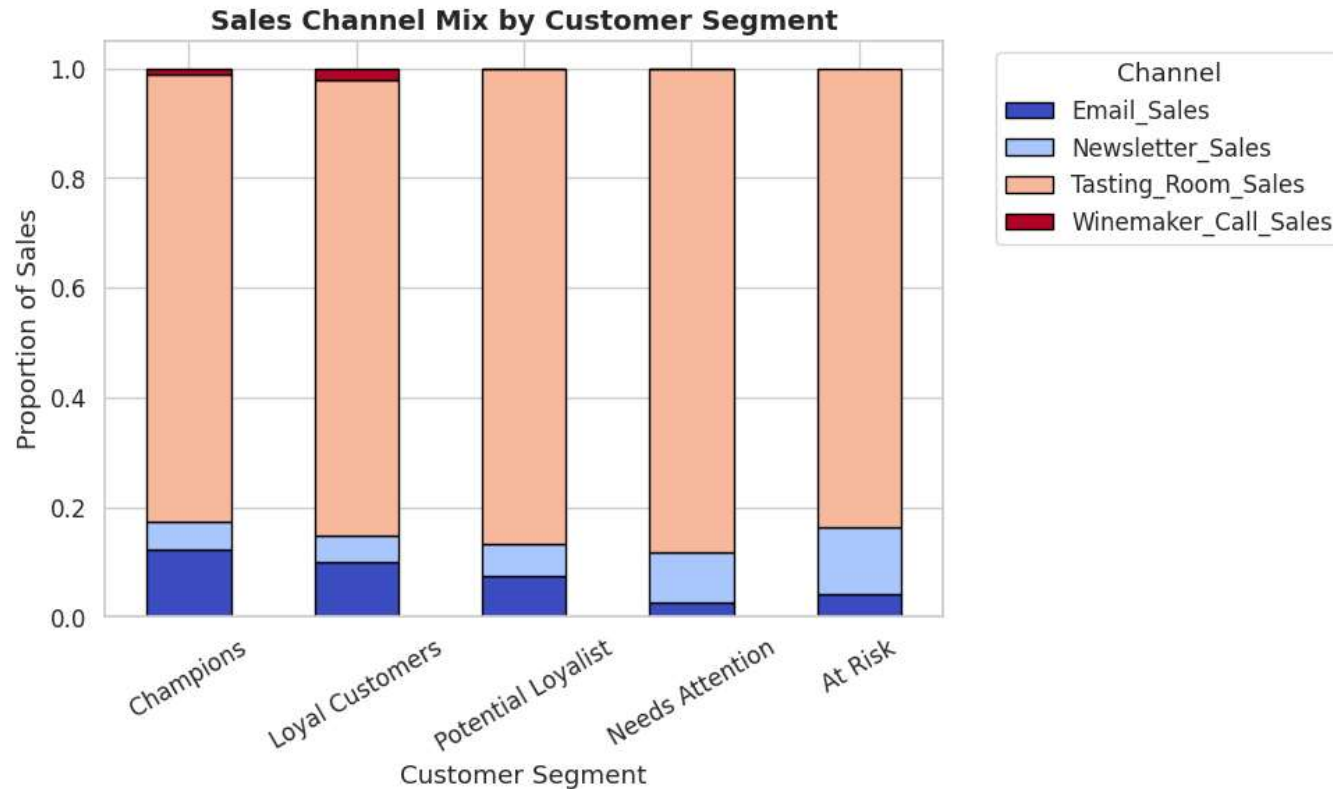


The RFM groups reveal **clear increases** in lifetime spending as customers move up the tiers

How can we progress customers into higher tiers?



Sales Mix by RFM Segment



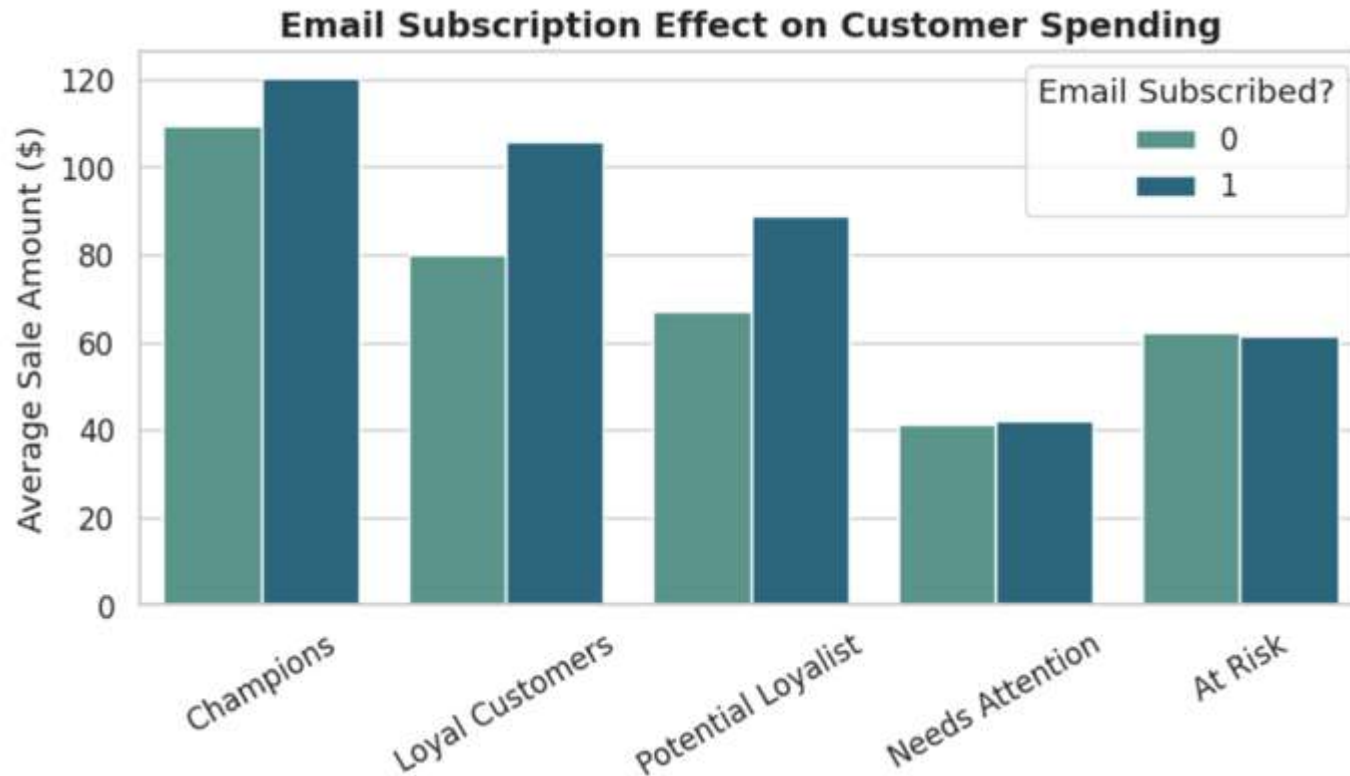
We see changes in the ways that customers spend as they move up tiers.

We see percentage of emails sales increases with tiers

We also see Winemaker Call sales only occur in the top two tiers



Average Spend of RFM Segment by Email Subscription

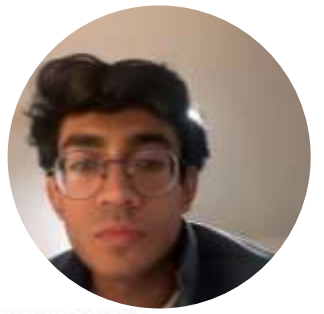


We see that email
Subscribers increase average sales
for only the top 3 tiers

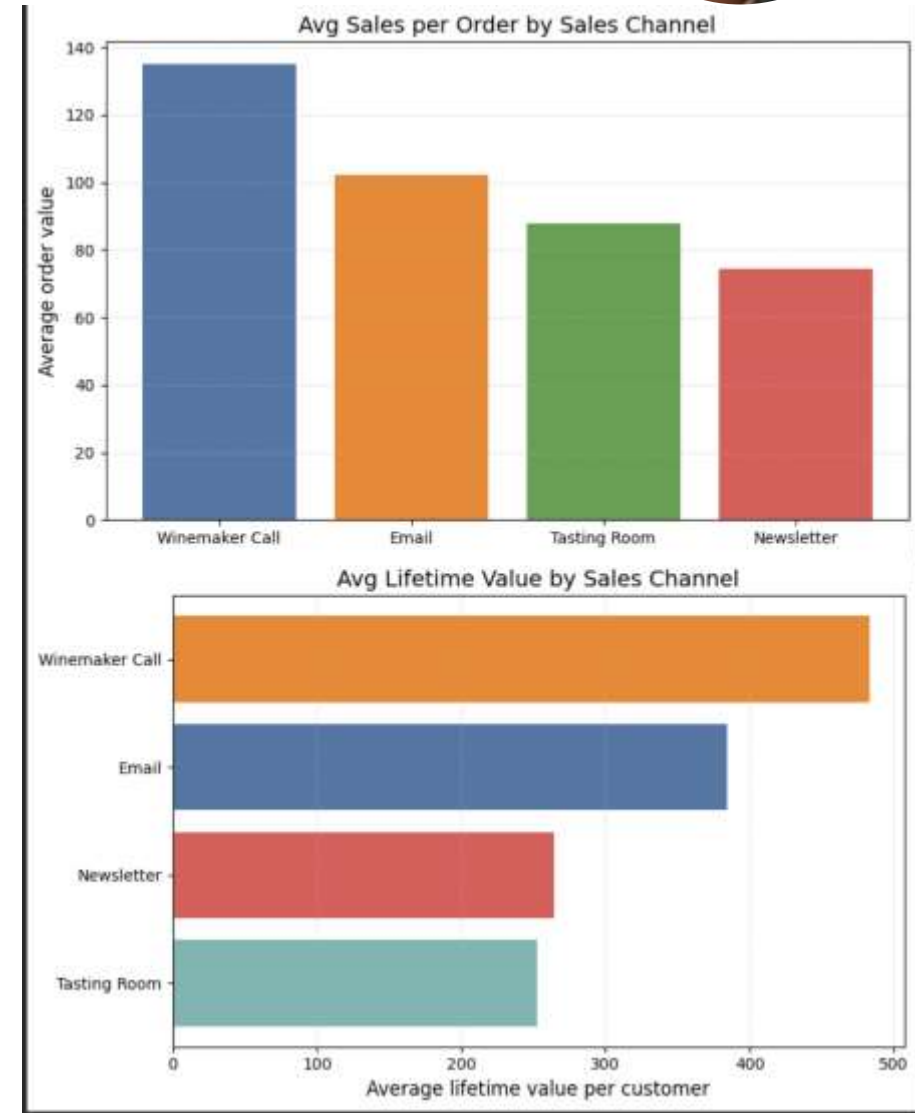
We should only target emails to
these top groups



Diving further – Winemaker Calls



- When we broke down performance by channel, Winemaker Calls had the **richest orders** and the same story held over time – Winemaker Call customers also delivered the **highest lifetime value**
- **Our path became obvious:** Let's focus on our highest margin channel aka the Winemaker Calls



Logistic Regression to figure out Customers to prioritize for Winemaker Calls



- Our question became: from what we know in 2008-2009, **who is most likely to buy via Winemaker Call in 2010?**
- A customer-level **logistic regression** model was built on pre-2010 spend, orders, segment and email/newsletter/tasting room activity
- The model worked well: **The top 10% of customers by score capture about 50% of all Winemaker Call buyers**

```
===== MODEL PERFORMANCE =====
... ROC AUC: 0.845

Confusion matrix (rows=actual, cols=predicted):
[[5094 1269]
 [ 34  85]]

Classification report:
              precision    recall  f1-score   support

     0       0.993      0.801      0.887     6363
     1       0.063      0.714      0.115      119

   accuracy      0.993
  macro avg       0.528      0.757      0.501     6482
 weighted avg     0.976      0.799      0.872     6482

Base Winemaker-Call buyer rate in TEST set: 0.0184

=== Lift vs base rate at different target sizes ===
Top 1% (n=64): buyer rate=0.156, lift=8.5x, captured buyers=10/119 (8.4% of all buyers in test)
Top 5% (n=324): buyer rate=0.117, lift=6.4x, captured buyers=38/119 (31.9% of all buyers in test)
Top 10% (n=648): buyer rate=0.093, lift=5.0x, captured buyers=60/119 (50.4% of all buyers in test)
Top 20% (n=1296): buyer rate=0.065, lift=3.5x, captured buyers=84/119 (70.6% of all buyers in test)

===== TOP 20 CUSTOMERS TO TARGET =====
      proba_win_2010  target_win_2010 Customer Segment Sales 2008 \
Customer ID
12              1.000              0  Casual Visitor      75.0
2779             0.998              0   High Roller       0.0
10030            0.998              0   High Roller       0.0
2733             0.998              0   High Roller       0.0
8296             0.998              0   High Roller       0.0
2627             0.997              0   High Roller       0.0
```

Turning Predictions Into a Targeted Call List



- **Top 20 customers ranked by predicted probability of a 2010 Winemaker Call purchase**, representing the highest priority prospects if calling capacity is very limited
- They are “lookalikes” of 2010 Winemaker Call buyers **based on their 2008–2009 behavior**, not just people who already used the channel
- This list lets us deploy scarce winemaker time where it matters most, **increasing expected high margin revenue per call**



Predicting Success: Who Says "Yes" to the Winemaker?

- 1. The "Green Light" (Prioritize Calls):
 - High Rollers
 - Recency
- 2. The "Red Light" (Stop Calling):
 - Email Subscribers
 - Luxury Estates
- 3. The Myth Buster:
 - Total Spend (Monetary)



Logit Regression Results						
Dep. Variable:	Target	No. Observations:	8796			
Model:	Logit	Df Residuals:	8787			
Method:	MLE	Df Model:	8			
Date:	Sat, 06 Dec 2025	Pseudo R-squ.:	0.3779			
Time:	02:40:36	Log-Likelihood:	-1149.9			
converged:	True	LL-Null:	-1848.3			
Covariance Type:	nonrobust	LLR p-value:	2.758e-296			
	coef	std err	z	P> z	[0.025	0.975]
const	0.4009	0.185	2.163	0.031	0.038	0.764
Recency	-0.0122	0.001	-16.762	0.000	-0.014	-0.011
Frequency	-0.0137	0.026	-0.525	0.600	-0.065	0.037
Monetary	-3.592e-05	3.08e-05	-1.167	0.243	-9.62e-05	2.44e-05
Email Subscr	-2.4143	0.161	-14.960	0.000	-2.731	-2.098
Newsletter Subscr	-0.5914	0.166	-3.569	0.000	-0.916	-0.267
Customer Segment_High Roller	1.7180	0.162	10.612	0.000	1.401	2.035
Customer Segment_Luxury Estate	-55.6403	1.23e+11	-4.51e-10	1.000	-2.42e+11	2.42e+11
Customer Segment_Wine Enthusiast	-0.0053	0.188	-0.028	0.978	-0.375	0.364

--- Clean Summary Table ---				
	Feature	Coefficient	P-Value	Odds Ratio
0	const	0.400853	3.052889e-02	1.493098e+00
1	Recency	-0.012151	4.639361e-63	9.879226e-01
2	Frequency	-0.013677	5.996163e-01	9.864165e-01
3	Monetary	-0.000036	2.431035e-01	9.999641e-01
4	Email Subscr	-2.414287	1.341606e-50	8.943107e-02
5	Newsletter Subscr	-0.591416	3.579469e-04	5.535430e-01
6	Customer Segment_High Roller	1.718003	2.628654e-26	5.573390e+00
7	Customer Segment_Luxury Estate	-55.640310	1.000000e+00	6.850474e-25
8	Customer Segment_Wine Enthusiast	-0.005260	9.777347e-01	9.947534e-01

Churn Test: Does Email Create Loyalty?

Logistic Regression for Churn prediction

1. The Cohort (Who):

Target Population: Customers acquired pre-2010 only

2. The Dependent Variable (What):

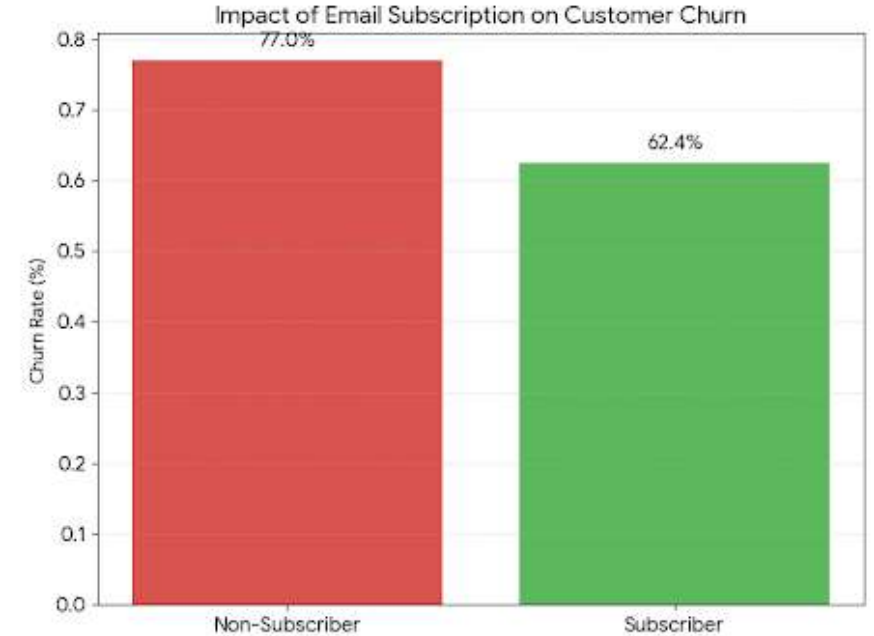
Churned (1): Customer spent \$0.00 in 2010

Retained (0): Customer made at least one purchase in 2010

3. The Controls (The "Equalizer"):

Variables Held Constant: Customer Segment (Wealth) and Past Spend (Value)

Objective: Isolate the pure impact of the Email Subscription by comparing customers of identical quality



Is this true??



The Retention Paradox: Is Our Email Driving Customers Away?

1. The "Shocking" Discovery (The Paradox):

Email Subscription: Positive Coefficient (+0.11).

Statistically Significant: (P-Value = 0.010).

Insight: Controlling for customer type, subscribers are ~12% MORE likely to churn than non-subscribers.

2. The Real Drivers of Loyalty (The Controls):

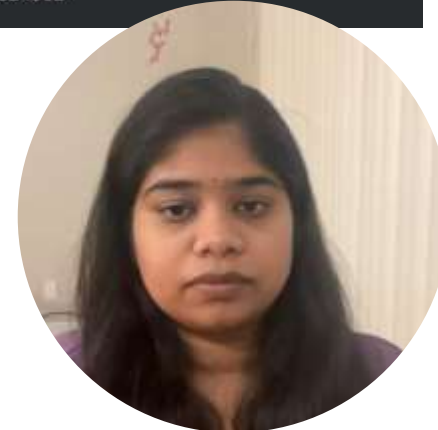
High Rollers: Coefficient -1.99 (Massively decreases churn).

Wine Enthusiasts: Coefficient -1.49 (Strongly decreases churn).

Insight: Our "loyal" customers stay because of who they are (Segments), not because of the email.

Logit Regression Results						
Dep. Variable:	Churned	No. Observations:	21046			
Model:	Logit	Df Residuals:	21040			
Method:	MLE	Df Model:	5			
Date:	Sat, 06 Dec 2025	Pseudo R-squ.:	0.08278			
Time:	19:17:52	Log-Likelihood:	-11371.			
converged:	True	LL-Null:	-12397.			
Covariance Type:	nonrobust	LLR p-value:	0.000			
	coef	std err	z	P> z	[0.025	0.975]
const	1.5874	0.027	58.187	0.000	1.534	1.641
Email Subscr	0.1113	0.043	2.589	0.010	0.027	0.196
Past_Spend	-2.431e-05	2.31e-05	-1.053	0.292	-6.96e-05	2.09e-05
Customer Segment_High Roller	-1.9893	0.057	-34.950	0.000	-2.101	-1.878
Customer Segment_Luxury Estate	-0.5158	0.041	-12.498	0.000	-0.597	-0.435
Customer Segment_Wine Enthusiast	-1.4921	0.052	-28.690	0.000	-1.594	-1.390

Clean Summary Table				
	Feature	Coefficient	P-Value	Odds Ratio
0	const	1.587401	0.000000e+00	4.891019
1	Email Subscr	0.111290	9.628526e-03	1.117719
2	Past_Spend	-0.000024	2.923396e-01	0.999976
3	Customer Segment_High Roller	-1.989308	1.298295e-267	0.136790
4	Customer Segment_Luxury Estate	-0.515781	7.630871e-36	0.597034
5	Customer Segment_Wine Enthusiast	-1.492089	5.077746e-181	0.224902



Correlation vs. Causation:

Does Multi-Channel Actually Drive Value?



Linear Regression

1. The Objective:

To prove that getting a customer to use a second channel (e.g., Tasting Room + Email) **causes** an increase in spending, rather than just correlating with it.

2. The Equation:

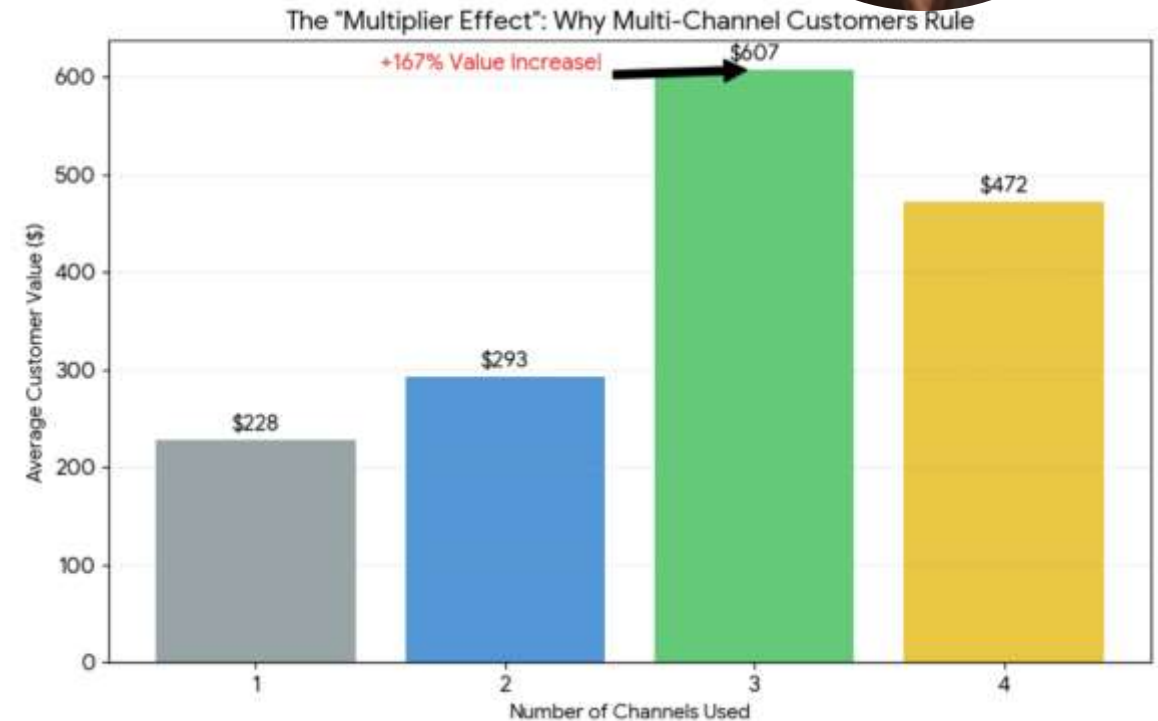
Target (Y): Customer Lifetime Value (Total Sales).

Primary Driver (X): Is_Multi-Channel (Binary: Uses 2+ Channels).

3. The Controls (Removing the Noise):

Tenure: We controlled for how long they have been a customer.

Wealth: We controlled for their Customer Segment (e.g., High Roller).



Multi-Channel Customer
or
Single-channel high value Customer?

Linear Regression: Multi-Channel Causation

Results:

1. The Omnichannel Premium: +\$44.62 per customer

Significance: Highly Significant (P-Value = 0.006)

2. The "Time" Comparison:

Value of 1 Year Loyalty: ~\$15 (Tenure_Years Coeff)

Value of 1 Second Channel: ~\$45

Insight: 1 Channel Conversion = 3 Years of Retention

Interpretation: Getting a single-channel customer to use a second channel instantly adds ~\$45 to their value, independent of how long they stay.

OLS Regression Results						
Dep. Variable:	Sale Amount	R-squared:	0.025			
Model:	OLS	Adj. R-squared:	0.025			
Method:	Least Squares	F-statistic:	116.0			
Date:	Sat, 06 Dec 2025	Prob (F-statistic):	1.54e-121			
Time:	03:39:51	Log-Likelihood:	-1.8934e+05			
No. Observations:	22869	AIC:	3.787e+05			
Df Residuals:	22863	BIC:	3.787e+05			
Df Model:	5					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	140.5190	10.933	12.853	0.000	119.090	161.948
Is_Multi_Channel	44.6243	16.207	2.753	0.006	12.857	76.391
Tenure_Years	14.6890	5.294	2.774	0.006	4.312	25.066
Customer Segment_High Roller	479.2455	21.674	22.111	0.000	436.763	521.728
Customer Segment_Luxury Estate	42.7743	16.940	2.525	0.012	9.571	75.977
Customer Segment_Wine Enthusiast	100.9192	19.998	5.047	0.000	61.723	140.116
Omnibus:	48920.291	Durbin-Watson:	0.138			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	283766870.543			
Skew:	18.932	Prob(JB):	0.00			
Kurtosis:	547.396	Cond. No.	8.52			



Recommendations

Continue leveraging Tasting Room as the anchor channel

- Highest-value driver of sales and engagement
- Host regular events to drive engagement
- Utilize other channels to encourage in-person visits

Invest in Winemaker Calls

- Invest in loyal customers with high growth potential
- Deepen relationships and encourage larger purchases

Revamp Email channel strategy

- Personalized by segment and customer journey stage
- Streamlined and supportive of other channels
- Limit frequency to reduce churn and annoyance

Drive more multi-channel engagement

- Integrated campaigns reinforce engagement and loyalty



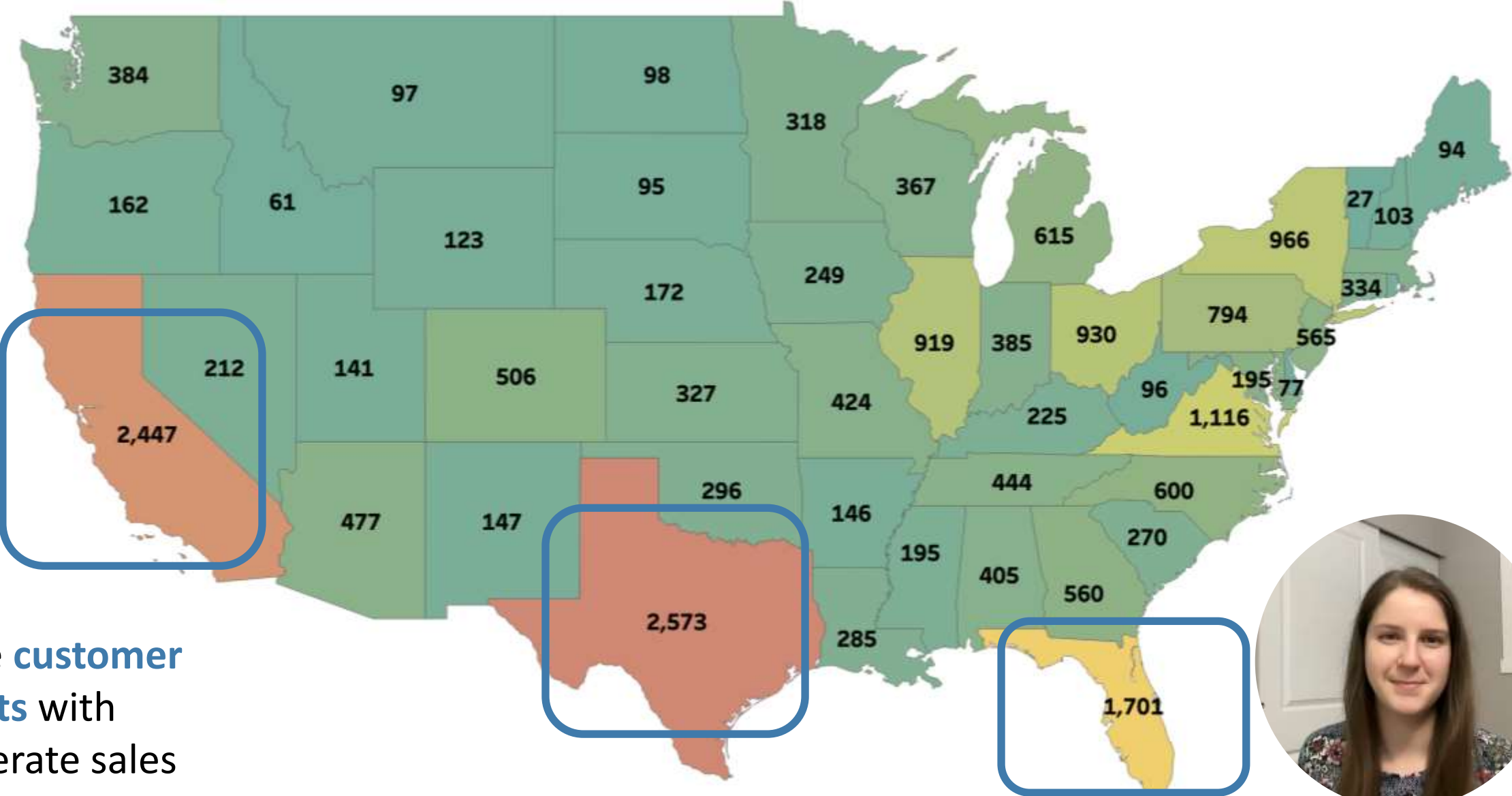
Future Analysis: Geo-Target by Zip Code

Analyze customers zip codes to implement geo-targeted outreach campaigns, sharpening marketing precision and maximizing customer value

- Reinforce loyalty in high-performing areas
- Convert high-potential regions into growth markets
- Target new lookalike customers
- Personalize campaigns for each region
- Drive immediate sales as well as long-term expansion



Distinct Customers by State



Large **customer counts** with moderate sales



Thank you!

